AFDB · DATA INNOVATION FOR AFRICA (DI4A) · AFRICA AFTER DARK

Night-Time Lights

What the darkness tells us — reading Africa's development from space



Concepts, instruments, artefacts and the workflow behind four hands-on laboratories: from raw radiance to a validated subnational indicator.

Data Innovation For Africa (DI4A) - AfDB

SECTION 1

01

Foundations

Why should a photograph of the dark tell us anything at all?

The core idea · Definitions · The inferential chain



Light is what human activity leaks

Almost everything people do after sunset requires light — and part of that light escapes upward, into the field of view of a satellite.

That physical fact turns the night side of the planet into an observation of human presence that no census, survey or administrative register can match on all four properties at once:

Globally consistent
Same method in every country

Spatially continuous
Every ≈500 m grid cell

Politically neutral
Independent of reporting

Freely available
Open NASA / EOG products

1992
Start of the continuous digital DMSP record — three decades of history

500 m
VIIRS Day/Night Band pixel, nearly constant across the swath, data product

Nightly
Global VIIRS overpass around 01:30 local time, since 2012

For an NSO: censuses come roughly once a decade, surveys every few years — lights arrive every night, for every district.

What night-time lights are — and are not

WORKING DEFINITION

Night-time lights (NTL) are measurements of visible and near-infrared radiance travelling upward from the Earth's surface, recorded by a satellite sensor during local night, when the scene is free of direct sunlight.

What it is

A physical measurement
Radiance in $nW \cdot cm^{-2} \cdot sr^{-1}$ for calibrated sensors (VIIRS).

A consistent grid
Same method over every country — Rwanda and Nigeria alike.

A near-real-time record
Daily since 2012; monthly and annual composites, free of charge.

What it is not

A photograph
The familiar "Earth at night" is a statistical composite of many cloud-free nights.

A direct measure of people or income
Those are inferences — each needs its own justification.

A complete record of lighting
Indoor, shielded and sideways light never reaches the sensor.

From electricity to a pixel value: five links

1 — Energy & infrastructure

A grid, generator or fuel supply exists and is operating.

→

2 — Activity after dark

People use that energy for lighting: streets, homes, markets, industry.

→

3 — Upward emission

A fraction of the light escapes upward rather than being shielded.

→

4 — Atmospheric transfer

Photons cross the atmosphere, attenuated by aerosols, vapour and cloud.

→

5 — Sensor response

The detector integrates them over its band into a radiance value.

Where the chain breaks in African contexts — a dark pixel can mean …

- **Link 1** — Load-shedding or fuel shortage — people are there, power is not
- **Link 2–3** — Solar lanterns and small home systems too faint to detect
- **Link 4** — Rainy-season cloud, harmattan dust, biomass-burning smoke
- **Link 5** — Rural light below the sensor's detection floor

SECTION 2

02

The instruments

What exactly is doing the measuring, and what can it resolve?

History · DMSP versus VIIRS · Saturation and dynamic range



Fifty years of looking down at the dark

1970s — DMSP-OLS night imagery (film). City lights noticed as a by-product of cloud detection.

1992 — Start of the digital DMSP archive → "stable lights" annual series to 2013.

2011 — Suomi NPP launched with VIIRS: first sensor calibrated for night-time radiance.

2012 — VIIRS monthly cloud-free composites; NASA Black Marble record begins (Jan 2012).

2017 — NOAA-20 launched with a second VIIRS; Black Marble twin products from 2018.

2018 — Luojia-1-01 demonstrates ≈130 m night imaging from a small satellite.

2021 — SDGSAT-1 Glimmer imager: colour night imagery at 10–40 m.

2022 — NOAA-21 launched: a third VIIRS extends the record into the 2030s.

The irony: the first instrument was built to see moonlit clouds. City lights were noise — it took two decades for anyone to treat the noise as the signal.

Two sensors, two different sciences

	DMSP-OLS · historical	VIIRS Day/Night Band · modern
Period	1992 – 2013 (archive)	2012 – present (standard)
Radiometry	6-bit digital numbers 0–63, no on-board calibration	14-bit, calibrated radiance in nW·cm^{-2}·sr^{-1}
Resolution	≈2.7 km, gridded at 30 arc-seconds (≈1 km)	500 m, nearly constant across the swath
Saturation	City cores clip at 63	Wide dynamic range, three gain stages
Sensitivity	Misses faint, rural lighting	Detects far fainter light than DMSP
Products	Annual composites only	Daily, monthly and annual — open licence

Next generation: Luojia-1-01 (≈130 m), SDGSAT-1 Glimmer (10 m panchromatic, 40 m colour) and commercial constellations resolve streets and distinguish lamp types by colour.

Why the brightest places say the least

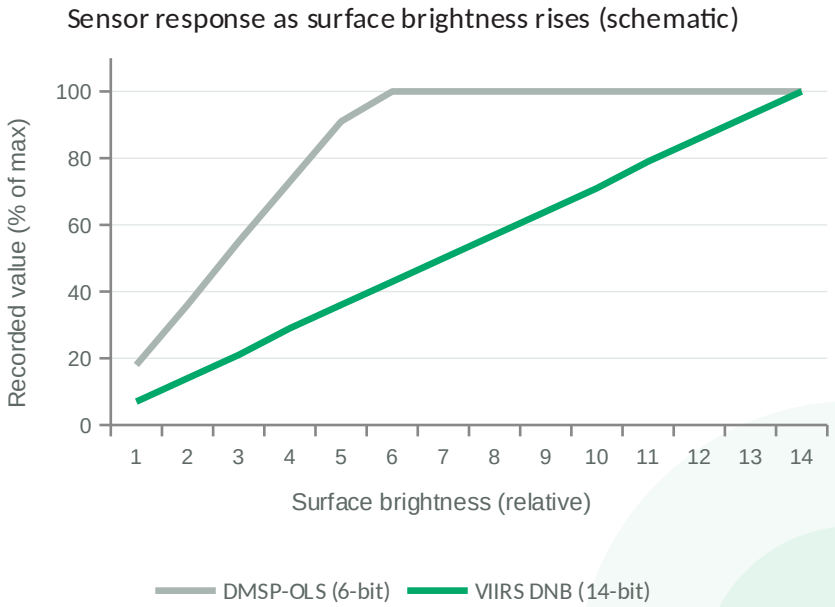
DMSP returns 63 above a certain brightness — a suburb and a downtown core look identical.

VIIRS spans roughly seven orders of magnitude, with three switchable gain stages.

Its low noise floor makes a single lit fishing boat visible on a dark lake or ocean.

Read this before you regress

Saturation is a ceiling, not noise you can average away. The information was never recorded — correcting it means modelling what the sensor failed to see.



Schematic illustration — not fitted data. VIIRS range per Elvidge et al. (2017).

SECTION 3

03

From photons to data

How does a night of observations become a number you can publish?

Processing · Products · What glows · Indicators · Example



How a night-lights image is manufactured

1. **Screen for cloud**
   Only nights flagged cloud-free enter a composite. In the wet tropics most nights may be discarded.
2. **Remove sunlight**
   Twilight, glint and instrument stray light are corrected or excluded.
3. **Correct for the Moon**
   Moonlight reflected by the ground mimics emitted light; it is modelled out.
4. **Filter transients**
   Fires, lightning, aurora and moving boats are flagged by their instability.
5. **Composite & tile**
   Surviving nights are averaged into a monthly or annual grid, tiled globally.

You will see these steps as layers in the Collect lab

Black Marble VNP46A2 carries the corrected radiance plus its evidence: cloud-mask flag, lunar irradiance, snow flag and a Mandatory_Quality_Flag (0 = high quality; 1–5 = poor, incl. aurora and glint).

A product, not a snapshot

Two products for the same month can legitimately differ. Always record product and version.

Sources: Román et al. (2018); NASA LAADS DAAC VNP46A2 v2 product page.

The NASA Black Marble product family

	VNP46A1 DAILY	VNP46A2 DAILY	VNP46A3 MONTHLY	VNP46A4 ANNUAL
Content	At-sensor radiance (top of atmosphere)	Moonlight-adjusted, gap-filled NTL	Composite of corrected daily NTL	Composite of corrected daily NTL
Corrections	Uncorrected — inputs & geometry only	Cloud, lunar, atmosphere, terrain, stray light	As A2, plus number of observations & quality	As A2, plus number of observations & quality
Typical use	Advanced users	Blackouts, reliability, events	Programme evaluation, seasonality	Long-run trends, subnational panels

Common to all: 15 arc-second grid (≈500 m at the equator) · 10°×10° tiles · HDF-EOS5 · from January 2012 · NOAA-20 twins named VJ146*.
Gotcha: radiance is stored as integers — multiply by the 0.1 scale factor and mask the 65535 fill value. **Alternatives:** EOG VIIRS Nighttime Lights (VNL) composites; Google Earth Engine catalogue.

What actually glows over Africa at night

Cities & settlements — SIGNAL

Streets, buildings, markets, vehicles. The dominant and most-studied part of the signal.

Corridors, ports & mines — SIGNAL

Highways, ports, airports, mining and industrial sites: the network between the bright spots.

Gas flares — MASK

Very bright, isolated points in oil & gas regions — e.g. the Niger Delta, Algeria, Libya. Mask them.

Fishing lights — CONTEXT

Lamp-attraction fishing on Lake Victoria and Lake Tanganyika; offshore fleets.

Fires — MASK

Savanna and agricultural burning: intense, short-lived and strongly seasonal.

Natural & reflected light — REMOVE

Moonlight on bright desert sand or salt pans, airglow. Not a human signal at all.

Gas flares are catalogued separately by the VIIRS Nightfire product (Earth Observation Group).

Turning an image into a number

Toy district · radiance per pixel

0.1	0.3	0.2	0.0
0.4	**2.5**	**6.0**	**0.6**
0.2	**4.0**	**12.0**	**1.5**
0.0	**0.8**	**1.2**	0.3

Shaded = above threshold 0.5 nW·cm^{-2}·sr^{-1}

30.1

Sum of Lights (SOL)

Total radiance inside a boundary. Workhorse of the economics literature; most sensitive to background noise.

8 px

Lit area

Pixels above a chosen threshold. Intuitive — but the threshold is a researcher choice.

3.6

Mean radiance (lit)

Brightness independent of extent. Separates a district getting brighter from one getting bigger.

Raise the threshold to 1.0 → lit area falls from 8 to 6 pixels (−25%). Always report product, version, boundary, threshold and masks — and test the threshold.

Illustrative values. Light concentration (Gini of radiance) is a fourth family, used for spatial inequality.

One question, end to end

Did the 2023 mini-grid programme actually bring light to its target villages?

1 State the claim in light terms

Electrified villages should show more nights above their background radiance after commissioning.

2 Choose the product

VNP46A3 monthly to date the change; VNP46A2 daily if you also want reliability. Record the version.

3 Define the units

Settlement footprints plus a small buffer — not whole districts, where a few lit villages vanish.

4 Pick indicator & control

Mean radiance and share of lit nights; controls = similar villages outside the programme.

5 Read the difference

A step change in programme villages that is absent in controls is evidence; a common rise is regional.

6 Try to break it

Check fires, flares, moon and season; compare with utility connection or programme monitoring records.

SECTION 4

04

Applications

Which statistical questions can this data actually answer?

SDG map · Economy · Energy access · Urbanisation · Crises



Where lights can support SDG monitoring

SDG 1

Poverty & wealth mapping

Small-area estimates combining lights with survey data

SDG 7

Electricity access & reliability

Locate unserved settlements; track how often lights are on

SDG 8

Economic activity

Subnational GDP downscaling; short-term activity signals

SDG 9

Infrastructure

Corridors, ports, industrial and special economic zones

SDG 11

Urbanisation

Urban extent, sprawl versus densification, new towns

SDG 16 · humanitarian

Conflict & disasters

Blackouts, displacement signals, recovery tracking

A complement, not a replacement. Lights add spatial detail, frequency and independent verification to official statistics — they do not replace the census or the survey.

Measuring the economy from orbit

What the evidence shows

Henderson, Storeygard & Weil (2012)

Lights carry real information on economic growth and can improve GDP estimates where national accounts are weak.

Bruederle & Hodler (2018)

Lights track household wealth, education and health at the local level, using DHS survey clusters in Africa.

Martinez (2022)

Lights used as an independent benchmark to test the credibility of reported GDP growth.

Why the lights–output link is not stable

- Sector mix: agriculture and mining create value with little light
- Income level: bright cores saturate or plateau
- Lighting technology: LED retrofits change light per watt
- Power supply: load-shedding dims lights, not output

WHERE LIGHTS HELP MOST

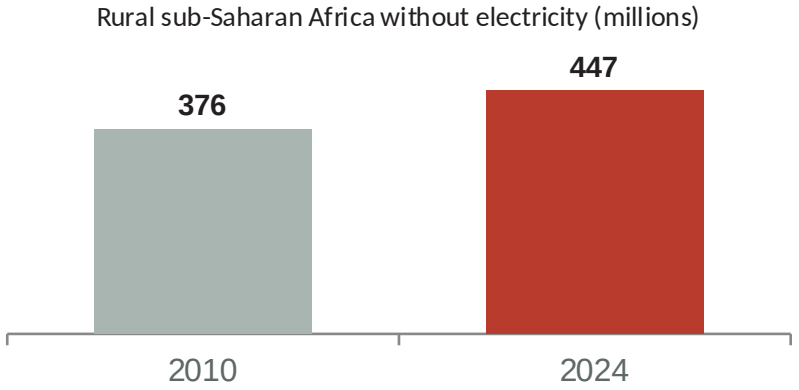
Subnational | High-frequency | Informal economy

Sources: Henderson et al. (2012) AER 102(2); Bruederle & Hodler (2018) PLOS ONE 13(9); Martinez (2022) JPE 130(10).

Energy access and SDG 7: Africa's defining gap

655 M
people without electricity worldwide (2024)

> 560 M
of them in sub-Saharan Africa



Locate the unserved
Settlements visible by day but dark at night are candidate unelectrified communities.

Verify programmes
A grid extension that produces no new light has not, in a meaningful sense, been delivered.

Measure reliability
Daily data reveal how often lights are on — load-shedding a binary "connected" statistic hides.

Plan the last mile
Distance to lit corridors informs grid extension vs mini-grid vs stand-alone solar.

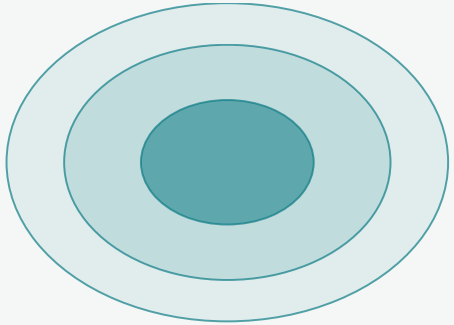
Mission 300 (World Bank Group & AfDB): 300 M people connected by 2030 — monitoring at that scale needs spatial evidence.

Sources: Tracking SDG 7: The Energy Progress Report (IEA, IRENA, UN DESA, World Bank, WHO), 24 June 2026; HREA, Min & O'Keeffe (Univ. of Michigan).

Urbanisation and the shape of a city

Sprawl

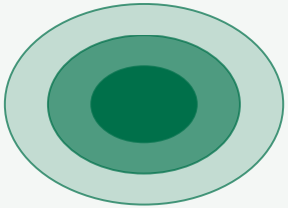
Footprint expands outward at low intensity



Lit area +40% · Radiance +6%

Densification

The same footprint intensifies



Lit area +6% · Radiance +40%

Same Sum of Lights — two very different cities (illustrative).

Map the footprint where registers are missing

Lights reveal built-up extent where cadastral records or recent maps do not exist.

Separate sprawl from densification

Compare growth in lit area with growth in radiance — it matters for transport and services.

Detect new towns early

Mining boomtowns, economic zones and displacement settlements appear before official geography.

Pair with population grids

Use WorldPop or GHSL — bright pixels are not people.

When the lights go out: conflict and disaster

Conflict · Sudan, 2023

After fighting began in Khartoum in April 2023, the capital's lights fell to the noise level by May (EOG). UNOSAT mapped power outages from NTL in June 2023.

Conflict · Syria, 2011–2014

Li & Li (2014) measured a loss of more than 70% of night-time light nationally, with losses tracking displacement province by province.

Disaster · Puerto Rico, 2017

After Hurricane Maria, NASA Black Marble daily data mapped the blackout and tracked restoration neighbourhood by neighbourhood.

The interpretive rule: absence of light is not absence of people

Darkness may mean displacement, destruction, fuel shortage, curfew, a grid failure — or a decision to switch off street lights. The satellite cannot tell which. Corroboration with ground information is not optional, and uncertainty must be stated.

Sources: Payne Institute / EOG (Sept 2024); UNOSAT (June 2023); Li & Li (2014) IJRS 35(18); Román et al. (2019) on Puerto Rico.

SECTION 5

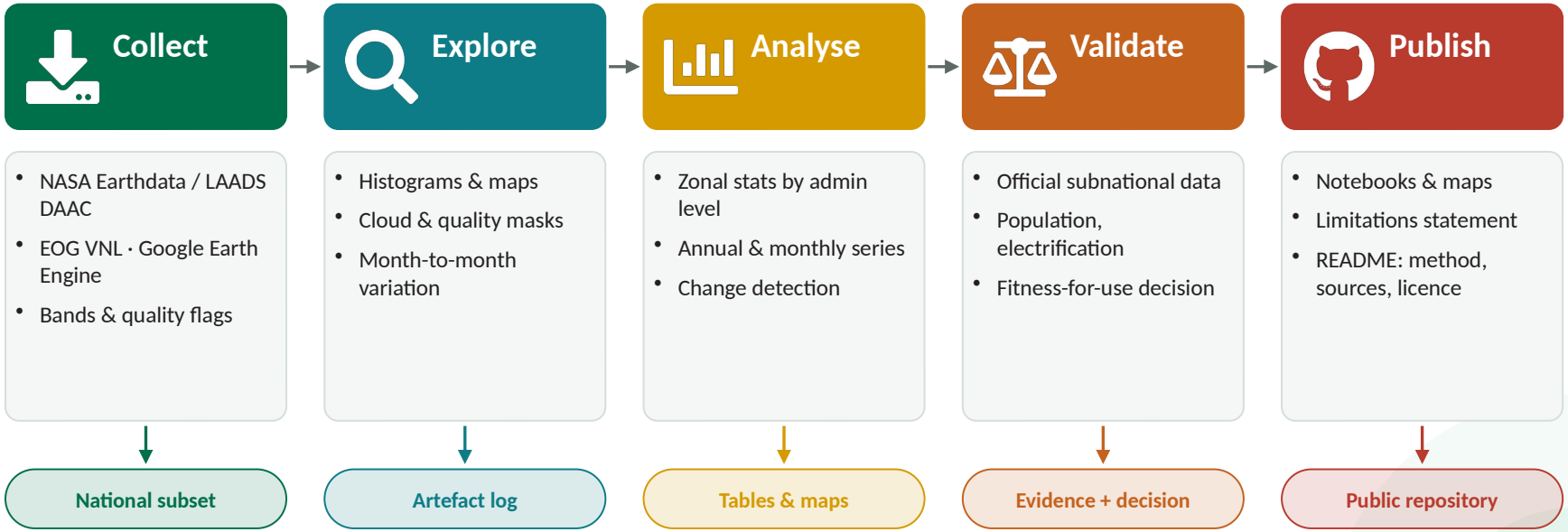
05

From talk to lab



Workflow · Choosing a product · Validation · Checklist · Discussion

Five stages, one reproducible pipeline



Example open-source stack: rasterio / xarray (read), geopandas (boundaries), rasterstats or exactextract (zonal statistics), matplotlib (maps).

Which cadence for which question?

Annual

VNP46A4 · EOG VNL annual

STRENGTH

Cleanest series; noise and ephemeral lights largely filtered.

WATCH OUT FOR

A change cannot be dated within the year.

TYPICAL NSO USE

Subnational panels, long-run trends, urban expansion, poverty mapping.

Monthly

VNP46A3 · EOG VNL monthly

STRENGTH

Dates a change to within a season.

WATCH OUT FOR

Rainy-season months may rest on very few valid nights.

TYPICAL NSO USE

Programme evaluation, seasonal activity, conflict & recovery timelines.

Daily

VNP46A2

STRENGTH

Every night, corrected for moon, atmosphere and terrain.

WATCH OUT FOR

Large volumes; a single night is noisy — aggregate first.

TYPICAL NSO USE

Blackouts, disaster response, reliability of electricity supply.

Rule of thumb: start annual. The finer the cadence, the more corrections you must trust and the more noise you must handle.

Publish, or keep as a diagnostic?

Q1 **Is the relationship strong and stable across units and years?**
Set your bar before you look.

↓

Q2 **Are artefacts identified and treated?**
Flares, blooming, seasonality, noise.

↓

Q3 **Is the benchmark independent?**
WorldPop uses NTL as a covariate — circular.

- **All three: yes →** Experimental statistic — publish with limitations
- **Partly →** Diagnostic tool — internal use only
- **No →** Not fit for purpose — document why

Limitations statement — template

Part of the deliverable, committed to GitHub

Data	Product, collection version, period, cadence
Choices	Boundaries, radiance threshold, masks applied
Artefacts	Which of the six were found, where, and how treated
Validation	Benchmark, year, number of units, strength of relationship
Decision	Experimental, diagnostic or not fit — and why
Must not be used for	e.g. household income, small-settlement counts

Suggested guide — adapt to your national quality assurance framework. Circularity: see Falchetta et al. (2019), Scientific Data.

A checklist for defensible analysis

1. **Name product and version**
   VNP46A4 v2, VNL V2.x — they are not interchangeable.
2. **Never splice sensors naively**
   Any DMSP–VIIRS series needs documented harmonisation.
3. **Report and test the threshold**
   Show the finding survives a reasonable range.
4. **Mask what you are not studying**
   Background noise, gas flares, fishing lights, fires.
5. **Normalise deliberately**
   Per capita, per km², or neither — state and defend it.
6. **Validate against the ground**
   Census, surveys, utility records — independent sources.
7. **Prefer changes to levels**
   Differences cancel much of the calibration error.
8. **Publish code and limitations**
   README with method, sources, licence and caveats.

THE GOLDEN RULE

State, in one sentence, what would have to be true about lighting behaviour for your conclusion to hold.

If you cannot write that sentence, you do not yet have a result.

Share it: a documented, reusable repository is how your office contributes to shared methodological guidance on new data sources.

IN CLOSING

Five things to take into the lab

1. Night lights are a proxy, never the quantity of interest — name the assumption that links them to your question.
2. The instrument defines the science: DMSP saturates, VIIRS calibrates, and neither sees blue light.
3. The data is most valuable where official statistics are thinnest: subnational, high-frequency, in crises and informal economies.
4. Darkness is not emptiness — off-grid light, cloud and load-shedding all look alike from orbit.
5. Validate against independent sources, and publish your limitations with your results.



The darkness tells us a great deal. It does not tell us everything — and it never speaks unprompted.

The vocabulary you will meet in the labs

Radiance
Energy leaving a surface per unit area and solid angle; for NTL in $nW \cdot cm^{-2} \cdot sr^{-1}$.

Digital number (DN)
Uncalibrated sensor count with no physical unit; DMSP reports 0–63.

Composite
Grid assembled from many cloud-free nights — not a photograph of one night.

Stable lights
Product with fires, aurora and other unstable sources removed.

Sum of Lights (SOL)
Total radiance summed over all pixels inside a boundary.

Blooming / overglow
Apparent spread of light beyond its source (scattering, geolocation error).

BRDF correction
Adjustment for how reflected (moon)light varies with viewing and lighting angles.

Zonal statistics
Summary of raster values (sum, mean, count) within each polygon.

Nadir
Point directly beneath the satellite, where the pixel footprint is smallest.

Skyglow
Diffuse brightening of the night sky by artificial light scattered in the atmosphere.

Selected references and data sources

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DATA & TOOLS

NASA Black Marble
LAADS DAAC · Earthdata

Earth Observation Group
VNL composites, Nightfire

Google Earth Engine
Hosted, analysis-ready archive

HREA
Settlement electricity access

World Bank Open Nighttime Lights
Tutorials & notebooks